IPM ASSIGNMENT 1

Good and Bad Interface Design

Rafaela Ferreira | 80004

7 October 2026 • 2026/2027

1. Introduction

This assignment examines one example of good interface design and one example of bad interface design. The analysis focuses on specific aspects of everyday interfaces and considers how mapping, feedback, recognition and the cost of user errors affect usability.

2. Good Interface Design

Example: The Dual-Flush Toilet Button

The aspect analysed is the flush control with two buttons, one for a half flush and one for a full flush. It is good design because the control itself tells you which button does what, and a wrong press costs almost nothing.

2.1. Objective of the Interface

A toilet flush control lets the user remove waste with water. In a dual-flush system it also lets the user choose how much water to use, so the interface must make that choice fast and obvious.

2.2. Good Design Aspects

- The two buttons have different sizes: a large one for the full flush and a small one for the half flush.
- Each button usually carries a symbol (a large and a small drop, or a full and a half circle), so no text is needed.
- The action takes one press, with no menu, no hold and no sequence.
- Feedback is immediate: the user hears and sees the water as soon as the button is pressed.
- The two buttons sit side by side on the same panel, so the choice is visible without searching.

2.3. Why It Is Good

- **Signifiers and mapping (Norman):** size and symbol communicate meaning. The bigger button means “more water”, which matches what the user expects.
- **Recognition rather than recall (Nielsen):** the user does not have to remember which button does what, because the answer is visible every time.
- **Visibility of system status:** the flush starts at once, so the user knows the command was received.

- **Error tolerance:** if the user picks the wrong button, the worst outcome is flushing again. There is no serious or irreversible consequence.
- **Efficiency and purpose:** one press serves both the common case and the economical case, and the design nudges people to use less water when a full flush is not needed.
- **Language independence:** it works for any user, including visitors who do not read the local language.

A limitation of this example: its quality depends on the implementation. Some models use two identical buttons or unclear symbols, which makes the choice a guess. Therefore, the good result comes from the size difference and clear symbols, not from having two buttons alone.



Figure 1 – Modern toilet flush



Figure 2 – Normal toilet flush

3. Bad Interface Design

Example: Stove Knobs That Do Not Match the Burners

The aspect analysed is the layout of the control knobs on a stove with four burners. It is bad design because the knobs sit in a straight row while the burners sit in a square, so the user cannot tell which knob controls which burner.

3.1. Objective of the Interface

A stove lets the user choose which burner to heat and how strongly, so they can cook safely. The controls must therefore make two things clear: which burner is selected and what its power level is.

3.2. Bad Design Aspects

- Four knobs are lined up in a row, while the four burners are arranged in two rows of two.
- The small symbols next to each knob are hard to read, especially from a standing position or in low light.

- The knobs give no clear sign of whether a burner is on, so the user must look at the burner or the flame.
- A user who is new to the stove, or a guest, cannot know the mapping without trying each knob.

3.3. Why It Is Bad

- **Poor natural mapping (Norman):** the position of the control does not correspond to the position of the thing it controls, so the user must learn or guess the link.
- **Recall instead of recognition (Nielsen):** the user has to remember which knob goes with which burner, which is easy to forget.
- **Error-prone design:** turning the wrong knob can heat an empty burner or leave the pot cold. With a heat source, errors can cause burns or fires, so the cost of a mistake is high.
- **Weak feedback:** there is no immediate sign on the knob of which burner is active.

3.4. Why It Was Probably Designed This Way

I believe the main reason is manufacturing. A single row of knobs fits one straight control panel, which is cheaper and simpler to build and reuse across models. The layout also looks neat, and it has become a common convention, so designers may have copied it without testing it with real users.

3.5. Suggested Improvements

1. Place each knob next to its burner, or arrange the four knobs in the same 2 by 2 pattern as the burners.
2. If a row is unavoidable, add a clear diagram next to each knob showing the burner it controls, with the active burner highlighted.
3. Use larger, high-contrast symbols that can be read from a standing position.
4. Add an indicator light, or a raised mark, to show when a burner is on.
5. Test the new layout with a few users, including people who have never used that stove, and check whether they pick the right burner on the first try.

3.6. Images of the Interface and Proposed Improvements



Figure 3 - Current Stove Interface

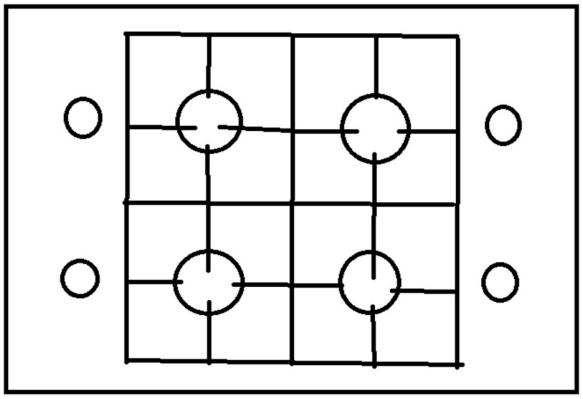


Figure 4 - Improved Proposement

4. Conclusion

Both examples come down to the same idea: a control should look like what it does. The flush button does this through size and symbols, so users choose correctly without thinking. The stove fails because the position of the knobs does not match the position of the burners even if it has the reference in each button, so users must learn or guess and it takes time. Neither interface is perfect, but these specific aspects show how mapping, feedback and error cost decide whether a design feels natural or frustrating.